Chapter 8. Inference in Multiple Regression II

Extra sums of squares, partial and sequential F, joint regions, simultaneous intervals, equality of regressions

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8.1 Extra sums of squares

Adding regressors increases SSR; the increase, the **extra sum of squares**, measures their contribution *given* those present.

For the full model $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$, write

$$SS(\hat\beta_0, \ldots, \hat\beta_p) = \hat\beta' X'\mathbf{y} \quad (df = p+1), \qquad SS(\hat\beta_0) = n\bar y^2 \quad (df = 1),$$

$$SS(\hat\beta_1, \ldots, \hat\beta_p \mid \hat\beta_0) = SS(\hat\beta_0, \ldots, \hat\beta_p) - SS(\hat\beta_0) = SSR \quad (df = p).$$

If a submodel uses only $x_1, \ldots, x_q$ ($q < p$) with fit $\tilde\beta$, the extra sum of squares of $x_{q+1}, \ldots, x_p$ given $x_1, \ldots, x_q$ is

$$SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q) = SS(\hat\beta) - SS(\tilde\beta) = SSR(\text{full}) - SSR(\text{sub}) = SSE(\text{sub}) - SSE(\text{full}),$$

with $p - q$ degrees of freedom. In matrix form, partition $\beta = (\beta_1', \beta_2')'$ with $\beta_1 = (\beta_0, \ldots, \beta_q)'$, $X = (X_1, X_2)$, and let $H_1 = X_1(X_1'X_1)^{-1}X_1'$, $H = X(X'X)^{-1}X'$:

$$SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H - H_1)\mathbf{y}.$$

Theorem 8.1

(a) $H - H_1$ is symmetric idempotent of rank $p - q$, and $(H - H_1)(I - H) = 0$.

(b) Under normal errors, $SS(\beta_2 \mid \beta_1)/\sigma^2 \sim \chi^2(p-q, \lambda)$ with $\lambda = \beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$, independent of SSE.

(c) Hence, under $H_0 : \beta_{q+1} = \cdots = \beta_p = 0$,

$$F_0 = \frac{SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q)/(p-q)}{MSE} \sim F(p-q, n-p-1).$$

Proof. (a) $HX = X$ gives $HX_1 = X_1$, hence $HH_1 = H_1$ and (transposing) $H_1H = H_1$. Then $(H-H_1)^2 = H - H_1 - H_1 + H_1 = H - H_1$, and $\operatorname{tr}(H - H_1) = (p+1) - (q+1)$. Also $(H - H_1)(I - H) = H - H - H_1 + H_1 = 0$.

(b) By Theorem 5.5(b) the noncentrality is $\frac{1}{2\sigma^2}(X\beta)'(H - H_1)X\beta$. Since $(H - H_1)X_1 = 0$ and $(H - H_1)X_2 = (I - H_1)X_2$, this equals $\beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$. Independence follows from (a) and Theorem 5.6.

(c) Under H_0, $\lambda = 0$. ∎

This is the same test as the general linear hypothesis with $C = (0, I_{p-q})$ (Chapter 7), since $SSE(\text{sub}) - SSE(\text{full}) = Q$.

Including the intercept in SSR gives the uncorrected table: $SSR = \hat\beta' X'\mathbf{y}$ ($p+1$ d.f.), $SST = \mathbf{y}'\mathbf{y}$ (n), and $MSR/MSE \sim F(p+1, n-p-1)$ tests $\beta = \mathbf{0}$.

8.2 Partial and sequential F tests

Partial F test. The contribution of a single regressor x_j given all others is $SS(\hat\beta_j \mid \text{all others})$, with one degree of freedom. The test of $H_0 : \beta_j = 0$,

$$F_0 = \frac{SS(\hat\beta_j \mid \hat\beta_0, \ldots, \hat\beta_{j-1}, \hat\beta_{j+1}, \ldots, \hat\beta_p)}{MSE} \sim F(1, n-p-1),$$

is the **partial F test**. It is equivalent to the t test, since

$$SS(\hat\beta_j \mid \text{others}) = \frac{\hat\beta_j^2}{c_{jj}}, \qquad F_0 = \left(\frac{\hat\beta_j}{\sqrt{c_{jj} MSE}}\right)^2 = t_0^2.$$

(Theorem 7.3 with $C = \mathbf{e}_j'$: $Q = \hat\beta_j^2/c_{jj}$.) A non-significant partial F means x_j adds little *given the others*, not that it is unrelated to y; with correlated regressors all partial Fs can be small while the overall F is large.

Sequential F tests. Enter regressors one at a time, testing each given those already in:

$$SS(\hat\beta_1 \mid \hat\beta_0), \quad SS(\hat\beta_2 \mid \hat\beta_0, \hat\beta_1), \quad \ldots, \quad SS(\hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_{p-1}).$$

These sum to SSR; each is compared with the stage (or full-model) MSE. The decomposition depends on order unless the centred regressors are orthogonal, and needs an unbiased MSE (omitted variables without effect).

8.3 Joint confidence regions and simultaneous intervals

Motivation. Joint regions, Bonferroni and Scheffé intervals control the error rate of all statements at once.

Theorem 8.2 (Joint confidence region)

Under normal errors,

$$\frac{(\hat\beta - \beta)' X'X (\hat\beta - \beta)}{(p+1) MSE} \sim F(p+1, n-p-1),$$

so $\{\beta : (\hat\beta - \beta)' X'X (\hat\beta - \beta) \le (p+1) MSE\, F_\alpha(p+1, n-p-1)\}$ is a $100(1-\alpha)\%$ confidence ellipsoid for β.

Proof. $\hat\beta - \beta = (X'X)^{-1}X'\epsilon$, so $(\hat\beta - \beta)' X'X (\hat\beta - \beta) = \epsilon' H \epsilon$, and $\epsilon' H \epsilon / \sigma^2 \sim \chi^2(p+1)$. It is independent of $SSE = \epsilon'(I - H)\epsilon$, since $H(I - H) = 0$. ■

Separate 95% intervals for $\beta_0, \ldots, \beta_p$ do not have joint coverage 95%. Two remedies:

Theorem 8.3 (Bonferroni and Scheffé intervals)

(a) (Bonferroni) The intervals $\hat\beta_j \pm t_{\alpha/(2(p+1))}(n-p-1)\sqrt{c_{jj} MSE}$, $j = 0, \ldots, p$, cover all β_j simultaneously with probability at least $1 - \alpha$.

(b) (Scheffé) With probability $1 - \alpha$, simultaneously for all $\mathbf{a} \in \mathbb{R}^{p+1}$,

$$\mathbf{a}'\beta \in \mathbf{a}'\hat\beta \pm \sqrt{(p+1) F_\alpha(p+1, n-p-1)} \sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}.$$

Proof. (a) If A_j is the event that the j-th interval covers, $P(A_j^c) = \alpha/(p+1)$, and $P(\bigcap A_j) \ge 1 - \sum P(A_j^c) = 1 - \alpha$.

(b) By the Cauchy–Schwarz inequality in the inner product defined by $(X'X)^{-1}$,

$$\max_{\mathbf{a} \neq \mathbf{0}} \frac{\{\mathbf{a}'(\hat\beta - \beta)\}^2}{\mathbf{a}'(X'X)^{-1}\mathbf{a}} = (\hat\beta - \beta)' X'X (\hat\beta - \beta),$$

so the event in (b) is exactly the joint confidence region of Theorem 8.2. ■

$\mathbf{a} = \mathbf{x} = (1, x_1, \ldots, x_p)'$ in (b) gives a **confidence band** for the whole surface. Bonferroni is shorter for few coefficients; Scheffé covers infinitely many combinations.

8.4 Testing equality of regression models

Two populations follow

$$\mathbf{y}_1 = X_1\beta_1 + \epsilon_1, \qquad \mathbf{y}_2 = X_2\beta_2 + \epsilon_2,$$

with n_1 and n_2 observations, design matrices of size $n_i \times (p+1)$, and independent $N(\mathbf{0}, \sigma^2 I)$ errors with a *common* σ^2. To test $H_0 : \beta_1 = \beta_2$:

(1) Full model: fit each separately, $SSE(F) = SSE_1 + SSE_2$, with $n-2(p+1)$ degrees of freedom ($n = n_1 + n_2$).

(2) Reduced model: stack $\mathbf{y} = (\mathbf{y}_1', \mathbf{y}_2')'$ and $X = (X_1', X_2')'$ and fit one β_0: $SSE(R)$, with $n-(p+1)$ degrees of freedom.

(3) Under H_0,

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/(p+1)}{SSE(F)/(n-2p-2)} \sim F(p+1, n-2p-2).$$

The reduced model is the full model (design $\mathrm{diag}(X_1, X_2)$, parameter $(\beta_1', \beta_2')'$) under $p+1$ restrictions $\beta_1 - \beta_2 = \mathbf{0}$: a general linear hypothesis. For k groups the d.f. are $(k-1)(p+1)$ and $n - k(p+1)$.

The test assumes equal error variances; with $k = 2$ this can be checked first by $F = MSE_1/MSE_2 \sim F(n_1 - p - 1, n_2 - p - 1)$.

8.5 Selected exercises

Motivation. The equality test via coefficient differences, partial F through R^2, order-free sequential sums, and Scheffé vs t intervals.

Exercise 8.1 (Equality of two regressions via $\hat\beta_1 - \hat\beta_2$)

In the setting of Section 8.4, let $\hat\beta_1, \hat\beta_2$ be the separate least squares estimators, $G_i = (X_i'X_i)^{-1}$, $\hat\Delta = \hat\beta_1 - \hat\beta_2$ and $MSE = (SSE_1 + SSE_2)/(n-2p-2)$.
(a) Find the distribution of $\hat\Delta$, and show that $(n-2p-2)MSE/\sigma^2 \sim \chi^2(n-2p-2)$, independent of $\hat\Delta$.
(b) Show that under $H_0 : \beta_1 = \beta_2$,

$$F_1 = \frac{\hat\Delta'(G_1+G_2)^{-1}\hat\Delta/(p+1)}{MSE} \sim F(p+1, n-2p-2).$$

(c) Show that F_1 equals the statistic F_0 of Section 8.4.

Solution.

(a) $\hat\beta_i \sim N_{p+1}(\beta_i, \sigma^2 G_i)$ (Theorem 6.3), independent across samples, so $\hat\Delta \sim N_{p+1}(\beta_1 - \beta_2, \sigma^2(G_1 + G_2))$; by Theorem 6.4, $\hat\beta_1, SSE_1, \hat\beta_2, SSE_2$ are mutually independent, so $(SSE_1 + SSE_2)/\sigma^2 \sim \chi^2(n-2p-2)$ independent of $\hat\Delta$.

(b) $\Sigma = \sigma^2(G_1+G_2)$ is positive definite. Under H_0, $\hat\Delta \sim N(\mathbf{0}, \Sigma)$, and with $A = \Sigma^{-1}$ we have $A\Sigma = I$, which is idempotent of rank $p+1$; Theorem 5.5(a) gives $\hat\Delta'\Sigma^{-1}\hat\Delta \sim \chi^2(p+1)$. Dividing by $(n-2p-2)MSE/\sigma^2$ and using (a),

$$F_1 = \frac{\hat\Delta'\Sigma^{-1}\hat\Delta/(p+1)}{\{(n-2p-2)MSE/\sigma^2\}/(n-2p-2)} \sim F(p+1, n-2p-2).$$

(c) Write the full model as one regression with design $X_F = \operatorname{diag}(X_1, X_2)$ and parameter $\mathbf{b} = (\beta_1', \beta_2')'$. Then $X_F'X_F = \operatorname{diag}(X_1'X_1, X_2'X_2)$, so the normal equations decouple, $\hat{\mathbf{b}} = (\hat\beta_1', \hat\beta_2')'$ and $SSE(F) = SSE_1 + SSE_2$. The hypothesis is $C\mathbf{b} = \mathbf{0}$ with $C = (I_{p+1}, -I_{p+1})$, and

$$C(X_F'X_F)^{-1}C' = G_1 + G_2, \qquad C\hat{\mathbf{b}} = \hat\Delta, \qquad Q = \hat\Delta'(G_1+G_2)^{-1}\hat\Delta.$$

By Theorem 7.3 (Method II), $SSE(R) - SSE(F) = Q$, so $F_0 = \{Q/(p+1)\}/MSE = F_1$. ■

Exercise 8.2 (Extra sums of squares through R^2)

Let the full model (with intercept) have p regressors and coefficient of determination R_F^2, and the submodel the first q of them, with R_R^2.
(a) Show that the F statistic of Theorem 8.1(c) equals

$$F_0 = \frac{(R_F^2 - R_R^2)/(p-q)}{(1-R_F^2)/(n-p-1)}.$$

(b) Show that $R_F^2 \geq R_R^2$, with equality if and only if $\hat\beta_{q+1} = \cdots = \hat\beta_p = 0$ in the full fit.

Solution [Supplement].

(a) Both models contain the intercept and are fitted to the same $\mathbf{y}$, so they share SST, and $SSE = SST(1-R^2)$ for each. Hence $SS(\beta_2 \mid \beta_1) = SSE(\text{sub}) - SSE(\text{full}) = SST(R_F^2 - R_R^2)$ and $MSE = SST(1-R_F^2)/(n-p-1)$; SST cancels in the ratio.

(b) $R_F^2 - R_R^2 = \mathbf{y}'(H - H_1)\mathbf{y}/SST = \|(H-H_1)\mathbf{y}\|^2/SST \geq 0$, because $H - H_1$ is symmetric idempotent (Theorem 8.1(a)). By Theorem 6.5, the full-model residual is $(I-H)\mathbf{y} = (I-H_1)\mathbf{y} - X_{2\cdot1}\hat\beta_2$, so

$$(H-H_1)\mathbf{y} = (I-H_1)\mathbf{y} - (I-H)\mathbf{y} = X_{2\cdot1}\hat\beta_2.$$

$X_{2\cdot1} = (I-H_1)X_2$ has full column rank: if $X_{2\cdot1}\mathbf{a} = \mathbf{0}$, then $X_2\mathbf{a} = H_1X_2\mathbf{a}$ lies in the column space of X_1, contradicting the full rank of X unless $\mathbf{a} = \mathbf{0}$. Therefore $R_F^2 = R_R^2$ iff $X_{2\cdot1}\hat\beta_2 = \mathbf{0}$ iff $\hat\beta_2 = \mathbf{0}$. ■

Exercise 8.3 (When the order of entry does not matter)

Partition $X = (X_1, X_2)$ as in Section 8.1, X_1 containing the intercept, and suppose $X_1'X_2 = 0$ (for instance, the columns of X_2 are centred and uncorrelated with those of X_1). Let $H_2 = X_2(X_2'X_2)^{-1}X_2'$.
(a) Show that $H = H_1 + H_2$, and hence $SS(\beta_2 \mid \beta_1) = \mathbf{y}'H_2\mathbf{y}$, which does not involve X_1.
(b) Show that $\hat\beta_2 = (X_2'X_2)^{-1}X_2'\mathbf{y}$.
(c) Deduce that sequential sums of squares of mutually orthogonal centred blocks do not depend on the order.

Solution [Supplement].

(a) $H_1H_2 = X_1(X_1'X_1)^{-1}(X_1'X_2)(X_2'X_2)^{-1}X_2' = 0 = H_2H_1$, so $P = H_1 + H_2$ is an orthogonal projection whose range contains X_1, X_2 and lies in $\operatorname{col}(X)$: $P = H$, and $SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H-H_1)\mathbf{y} = \mathbf{y}'H_2\mathbf{y}$.

(b) $(I-H_1)X_2 = X_2 - X_1(X_1'X_1)^{-1}X_1'X_2 = X_2$, so $X_{2\cdot1} = X_2$, and by Theorem 6.5, $\hat\beta_2 = (X_2'X_2)^{-1}X_2'(I-H_1)\mathbf{y} = (X_2'X_2)^{-1}X_2'\mathbf{y}$, using $X_2'H_1 = 0$.

(c) For centred blocks with $Z_j'Z_k = 0$ and $\mathbf{1}'Z_j = \mathbf{0}$, earlier blocks plus intercept form an X_1 orthogonal to Z_j, so by (a) its sequential sum is $\mathbf{y}'H_{Z_j}\mathbf{y}$ in every order; these sum to SSR. ■

Exercise 8.4 (Scheffé intervals are wider than t intervals)

Let $\nu = n - p - 1$. Show that $(p+1)F_\alpha(p+1,\nu) \geq t_{\alpha/2}(\nu)^2$, so that for a single combination $\mathbf{a}'\beta$ the Scheffé interval of Theorem 8.3(b) contains the t interval of Theorem 7.2(b).

Solution [Supplement].

Step 1. Fix $\mathbf{a} \neq \mathbf{0}$ and let $T_{\mathbf{a}} = \mathbf{a}'(\hat\beta - \beta)/\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\,MSE}$. By Theorem 7.2, $T_{\mathbf{a}} \sim t(\nu)$, so $T_{\mathbf{a}}^2 \sim F(1,\nu)$, whose distribution function is continuous and strictly increasing on $(0,\infty)$.

Step 2. By the proof of Theorem 8.3(b), $\sup_{\mathbf{b}\neq\mathbf{0}} T_{\mathbf{b}}^2 = (\hat\beta-\beta)'X'X(\hat\beta-\beta)/MSE$, and by Theorem 8.2 this is at most $c = (p+1)F_\alpha(p+1,\nu)$ with probability $1-\alpha$.

Step 3. Since $T_{\mathbf{a}}^2 \leq \sup_{\mathbf{b}} T_{\mathbf{b}}^2$, $P(T_{\mathbf{a}}^2 \leq c) \geq 1-\alpha = P(T_{\mathbf{a}}^2 \leq t_{\alpha/2}(\nu)^2)$. By the strict monotonicity in Step 1, $c \geq t_{\alpha/2}(\nu)^2$. The half-widths are $\sqrt{c}$ and $t_{\alpha/2}(\nu)$ times the same standard error, which proves the claim. ■